1. (a) $\varepsilon_1 = \dfrac{(y+z) - fl(y+z)}{y+z}$ — (1/2)

$\varepsilon_2 = \dfrac{x\, fl(y+z) - fl(x\, fl(y+z))}{x\, fl(y+z)}$ — (1/2)

Recall: $E_r = \dfrac{x - fl(x)}{x} \Rightarrow \underbrace{fl(x) = x(1 - E_r)}_{(1/2)}$

We have

$fl(y+z) = (y+z)(1-\varepsilon_1)$ — (1/2)

$$\left.\begin{aligned} fl(x\, fl(y+z)) &= x(y+z)(1-\varepsilon_1)(1-\varepsilon_2) \\ &= x(y+z)(1 - \varepsilon_1 - \varepsilon_2 + \varepsilon_1\varepsilon_2) \\ &= x(y+z)(1 - (\varepsilon_1 + \varepsilon_2 - \varepsilon_1\varepsilon_2) \\ \Rightarrow E_r(x(y+z)) &= \frac{x(y+z) - fl(x\, fl(y+z))}{x(y+z)} \\ &= \varepsilon_1 + \varepsilon_2 - \varepsilon_1\varepsilon_2 \end{aligned}\right\} (1)$$

1.(b) $E_a := |f(x) - f(x_A)| \le f'(\xi)\, |x - x_A|$ — (1/2)

$$\left.\begin{aligned} |x - 2.5| &\le 0.01 \\ 2.49 \le x &\le 2.51 \end{aligned}\right\} (1/2)$$

$$\left.\begin{aligned} E_a &\le 3(2.51)^2 \times 0.01 \\ &= 0.189003 \end{aligned}\right\} — (1)$$

Using $\xi = 2.5$ is also fine in the third step but (1/2) marks for step 2 will be deducted in that case.

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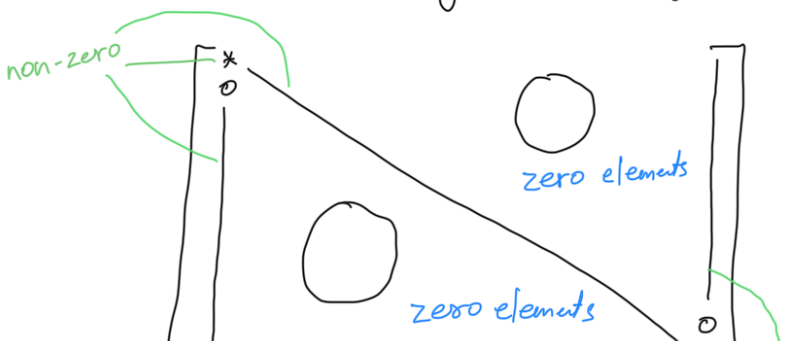
2. The structure of the given matrix



— (2) For identifying the matrix structure correctly.

$\times$ non-zero

Only diagonal, and first & last Column will have non-zero enteries.

<u>Elimination phase</u> : <u>Step 1</u>

multipliers $m_{i1} = \dfrac{a_{i1}}{a_{11}}$, $i = 3, \dots n$ — (1/2)

total number of divisions $= n-2$ — (1)
" " " multiplication $= n-2$ — (1)
" " " add \ sub $= n-2$ — (1)

total operation count $= 3(n-2)$ — (1/2)

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(3) <u>Recall</u> : Jacobi method converges if the matrix A in the system $Ax = b$ is diagonally dominant. — (1 1/2)

[marks should be awarded if other condition of convergence is written i.e. $\|B\|_\infty < 1$]

the given matrix is not diagonally dominant. as $7 < |5| + |2|$ — (1 1/2)

[Same marks are given if $\|B\|_\infty > 1$ is shown.]

Rearrange the equations to make the resulting matrix diagonally dominant.

Equation $7x_1 - x_2 - 5x_3 = -3$ and $7x_1 + 5x_2 + x_3 = 1$ both are diagonally dominant for x_1.
$|7| > |-1| + |-5| = 6$ $\quad$ $|7| > |5| + |1| = 6$ — (1)

However, no equation has such property for x_2 and x_3 — (1)

We can not make an arrangement for which Jacobi method converges — (1)

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4. Fixed point iteration for

$$x^3 - x - 1 = 0$$

Formulation: $x = g(x)$

1. $g_1(x) = x^3 - 1$ — (1/2)

2. $g_2(x) = (x+1)^{1/3}$ — (1/2)

any two formulations are fine. Note that, more than 2 formulations are possible.

Convergence criteria.

1. $g(x)$ is self map
2. $g(x)$ is cont^s
3. $|g'(x)| < 1 \quad \forall\, x \in I$ (suitable interval).

$g_1(x) = x^3 - 1$ is self-map on $\mathbb{R}$, it is cont^s.

$g_1'(x) = 3x^2$ which is > 1 for any $|x| \geq 1$. (1)

with $g_1(x)$, fixed-pt iteration will diverge. — (1/2)

Similarly, $g_2'(x) = \dfrac{1}{3(x+1)^{2/3}}$ — (1)

For $x = 1.32$, $|g_2'(x)| \simeq 0.190204 < 1$ (converges) (1/2)

Iterations with $x_0 = 1.32$

$x_1 = (1.3+1)^{1/3} \simeq 1.320006$

$x_2 = (1.320006+1)^{1/3} \simeq 1.323812$

$x_3 \simeq 1.324535$

$x_4 \simeq 1.324673$ — (2)

Pick atleast one formulation for which fixed-pt converges. Any random initial guess is fine.

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5. Using $x^{(0)} = (0, .5, .25)$

<u>Iteration 1</u>

$y_1 = (0, 3.5, -0.75)$

$\mu_1 = -8.75$ (appr. eigenvalue) — 1

$x_1 = (0, -.4, 1)$ (appro. eigenvector)

Iteration 2

$y_2 = (0, -.4, 1)$

$\mu_2 = 1$ — 1

$x_2 = (0, -0.4, 1)$

Iteration 3, 4, 5 --- all are same. — 1/2

With $x^{(0)} = (1, .5, .25)$

Iteration 1

$y_1 = (3, -0.5, 7.25)$

$\mu_1 = 7.25$ — 1

$x_1 = (.41379, -.06897, 1)$

Iteration 2

$y_2 = (1.24138, -.06897, 2.65517)$

$\mu_2 = 2.65517$

$x_2 = (0.46753, -0.02598, 1)$ 1

Iteration 3

$y_3 = (1.40260, -.02597, 2.87013)$

$\mu_3 = 2.87013$

$x_3 = (0.48869, -.00905, 1)$

Iteration 4

$y_4 = (1.46606, -.00905, 2.95475)$

$\mu_4 = 2.95475$

$x_4 = (0.496172, -.00306, 1)$ 1

Iteration 5

$y_5 = (1.48852, -.00306, 2.98469)$

$\mu_5 = 2.98469$

$x_5 = (0.49872, -.001026, 1)$

Power method converges for both initial guesses.

1/2

Note, one need not explicitly need to write approximated eigenvalue & corresponding eigen-vector. as sequence $\{\mu_k\}$ and $\{x_k\}$ denotes that, respectively.